

# Manuscripts considered — ulms-highgrade-pulm-mets-er-pr-neg-q qvt

*Master inventory: every paper reviewed in this case*

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# Manuscripts considered — ulms-highgrade-pulm-mets-er-pr-neg-

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Master inventory: 25 clinical (22 included, 3 considered & excluded) + 37 pre-clinical (34 included, 3 considered & excluded) + 24 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

## —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** ivonescimab
- **Indication / model:** unresectable or metastatic leiomyosarcoma, 1-3 prior lines (or declined first-line standard of care) (2L+); none required
- **Design:** single-arm phase 2 of ivonescimab (PD-1 x VEGF bispecific antibody)
- **Effect size:** — — ORR (primary)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** REC-617
- **Indication / model:** metastatic or unresectable RB1-negative leiomyosarcoma after ≥1 prior systemic line (2L+); RB1-negative by IHC (0% expression) required
- **Design:** open-label phase 1b of REC-617, a selective CDK7 inhibitor
- **Effect size:** — — safety/RP2D (primary)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** brenetafusp
- **Indication / model:** metastatic/unresectable synovial sarcoma or myxoid round cell liposarcoma, HLA-A 02:01-positive (2L+); HLA-A02:01 genotype required; PRAME-directed (ImmTAC)
- **Design:** phase 2 of brenetafusp (PRAME x CD3 ImmTAC) in rare cancers (NCI)
- **Effect size:** — — ORR (primary)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Lefler/Basu Mallick (2025) — JCO Precis Oncol

**Reference:**PMID 40601892 · **Type:** preclinical · **Inclusion:** excluded — Pan-sarcoma expression-outlier analysis that lands on chordoma; leiomyosarcoma was not among the subtypes with CDK7 outlier overexpression, and the cell-line work is chordoma only. Reviewed as the closest thing to a sarcoma-wide CDK7 rationale, and it does not support one in LMS.

- **Intervention:** Pan-sarcoma CDK/cyclin outlier expression (CDK7 in chordoma)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Pan-sarcoma expression-outlier analysis that lands on chordoma; leiomyosarcoma was not among the subtypes with CDK7 outlier overexpression, and the cell-line work is chordoma only. Reviewed as the closest thing to a sarcoma-wide CDK7 rationale, and it does not support one in LMS.

## Guo/Li (2025) — Front Immunol

**Reference:**PMID 40948800 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Single-cell map of the metastatic uLMS microenvironment
- **Indication / model:** single-cell RNA sequencing of four metastatic lesions (pelvic cavity, rectum, peritoneum, bladder) from one treatment-naïve uterine leiomyosarcoma patient, against five normal myometrium samples; multiplex immunofluorescence validation and TCGA-SARC survival correlation
- **Design:** An immunosuppressed niche built from exhausted CD8 T cells (LAG3, HAVCR2, TIGIT), M2-polarised CD163+ macrophages and immature N2 neutrophils, with MIF-CD74/CD44 and CXCL8 signalling sustaining angiogenesis and macrophage polarisation.
- **n:** n=1 patient (4 metastatic sites); n=5 normal myometrium controls
- **Effect size:** The metastatic microenvironment was dominated by exhausted CD8 T cells and M2 macrophages rather than by any effector population, with naïve markers (CCR7, MAL) lost along the pseudotime trajectory. The macrophage-heavy composition is the same compartment trabectedin depletes.
- **Variance:** vs normal uterine myometrium; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** One patient. Descriptive, hypothesis-generating, and no therapy was given; it earns its place only because uLMS-specific microenvironment data at single-cell resolution are otherwise almost absent.

## Ceuppens/Breckpot (2025) — Eur J Nucl Med Mol Imaging

**Reference:**PMID 39237746 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Alpha versus beta FAP-targeted radionuclide therapy
- **Indication / model:** human-FAP-expressing lung tumour model in immunocompetent mice; FAP-targeting single-domain antibody 4AH29 labelled with 225Ac or 131I, with and without PD-L1 blockade
- **Design:** Alpha emission deposits high linear-energy-transfer dose over a few cell diameters while a beta emitter spreads a lower dose further; the report also links radionuclide therapy to PD-L1 upregulation, which supplies the rationale for adding checkpoint blockade.
- **Effect size:** moderate — Tumour uptake was high for both radionuclides, with 225Ac retention decaying far more slowly (2.1 to 1.7 %IA/g between 1 and 24 h) than 131I (3.5 to 0.9 %IA/g). Both improved survival over vehicle, PD-L1 rose in 225Ac-treated tumours, and the 225Ac-plus-PD-L1-blockade combination was synergistic.

- **Variance:** vs vehicle solution; PD-L1 blockade alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** FAP was engineered onto the tumour cells, so this tests direct antigen targeting rather than the stromal-to-tumour crossfire that FAP therapy in a sarcoma would depend on. Lung carcinoma model, different binder chemistry from either agent this patient could reach.

—/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** all-trans retinoic acid (ATRA) + cemiplimab
- **Indication / model:** advanced or metastatic leiomyosarcoma after standard-of-care chemotherapy (2L+); none required
- **Design:** single-arm phase 2 of all-trans retinoic acid + cemiplimab
- **Effect size:** — — ORR (primary)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** [225Ac]RTX-2358
- **Indication / model:** relapsed/refractory soft-tissue sarcoma, FAP-positive by FAPi PET (2L+); FAP expression demonstrated on [64Cu]LNTH-1363S FAPi PET — FAP never assessed in this patient; sarcomas are characteristically FAP-high
- **Design:** phase 1 dose escalation of [225Ac]RTX-2358 (FAP-targeted alpha radioligand) with FAPi PET selection gate
- **Effect size:** — — safety/RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** pegylated liposomal doxorubicin (maintenance)
- **Indication / model:** advanced soft-tissue sarcoma with disease control after 4-8 cycles of first-line anthracycline-based treatment (MELODY) (1L (maintenance)); none required; randomization within 8 weeks of last anthracycline dose
- **Design:** randomized phase 2: pegylated liposomal doxorubicin maintenance vs active surveillance
- **Effect size:** — — PFS (primary)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Pautier/Duffaud (2024) — N Engl J Med

Reference: [PMID 39231341](#) · Type: clinical · Inclusion: included

- **Intervention:** Doxorubicin + trabectedin, then trabectedin maintenance (LMS-04) — final OS
- **Indication / model:** metastatic or unresectable uterine or soft-tissue leiomyosarcoma, chemotherapy-naïve (1L); same 150-patient ITT population as the 2022 primary report; analyses adjusted for uterine vs soft-tissue origin and disease stage
- **Design:** final overall-survival analysis of the randomized phase 3 LMS-04 trial (median follow-up 55 months)
- **n:** 150
- **Effect size:** 33 months — median OS
- **Variance:** 95% CI 26–48
- **Toxicities:** Abstract reports only that adverse events and dose reductions were more frequent with doxorubicin-trabectedin; per-term combination-phase rates are on the 2022 primary-report row.
- **Case fit:** strong
- **Notes:** This is the OS readout that turns LMS-04 from a PFS result into a survival benefit, and the main reason the regimen was chosen for her. Toxicities left empty after the full fallback chain failed: abstract has no per-term data, NEJM has no PMC deposit, CT.gov NCT02997358 hasResults=false, publisher page returns 403.

## Sun/Italiano (2024) — Mol Cancer

Reference: [PMID 38374062](#) · Type: preclinical · Inclusion: excluded — Phase I/II clinical trial with paired on-treatment biopsies rather than a preclinical study. Logged because it is the human counterpart of the trabectedin macrophage literature: CD68+CD163+ macrophages fell in 9 of 28 biopsied sarcoma patients and the shift tracked with clinical benefit.

- **Intervention:** Metronomic trabectedin + low-dose cyclophosphamide (TARMIC)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Phase I/II clinical trial with paired on-treatment biopsies rather than a preclinical study. Logged because it is the human counterpart of the trabectedin macrophage literature: CD68+CD163+ macrophages fell in 9 of 28 biopsied sarcoma patients and the shift tracked with clinical benefit.

## Lynch/Pollack (2024) — BMC Cancer

Reference: [PMID 39478506](#) · Type: preclinical · Inclusion: included

- **Intervention:** B7-H3 expression across soft-tissue sarcoma subtypes
- **Indication / model:** immunohistochemistry on 153 soft-tissue sarcoma specimens spanning 15 subtypes, scored in tumour and in associated vasculature
- **Design:** B7-H3 (CD276) is a cell-surface immune-regulatory molecule with limited normal-tissue expression, which is what makes it usable as an ADC or CAR target rather than as a driver.
- **n:** n=153 specimens, 15 subtypes
- **Effect size:** B7-H3 was expressed in 97% of specimens and at high level in 69.2%, with vascular positivity in 94.7%. Expression was independent of prior treatment, tumour size, grade and patient age, and uncorrelated with PD-L1 or PD-1.
- **Variance:** vs not applicable — expression survey; PD-L1 and PD-1 status compared where previously assessed; translatability: high
- **Toxicities:** n/a (preclinical)

- **Case fit:** partial
- **Notes:** Prevalence only — no functional or therapeutic readout, and per-subtype numbers for leiomyosarcoma are not broken out in the abstract. High prior probability of a positive stain is not the same as a dependency, and vascular positivity raises a separate on-target question.

### —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** abemaciclib + gemcitabine
- **Indication / model:** advanced leiomyosarcoma or dedifferentiated liposarcoma (phase 2 may be systemic-treatment naive) (1L); intact Rb protein by IHC required (phase 2); prior gemcitabine not allowed
- **Design:** phase 1/2 of abemaciclib (CDK4/6 inhibitor) + gemcitabine
- **Effect size:** — — MTD/RP2D (phase 1); PFS (phase 2)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** zanzalintinib
- **Indication / model:** metastatic or unresectable leiomyosarcoma after ≥2 prior lines (cohort 1) (3L+); none required
- **Design:** phase 2 of zanzalintinib (multi-targeted TKI) with LMS-specific cohort
- **Effect size:** — — ORR/PFS (primary per cohort)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** letrozole
- **Indication / model:** newly diagnosed FIGO stage I uterine leiomyosarcoma, ER-positive (>10% IHC), adjuvant setting (adj); ER expression >10% by IHC required
- **Design:** randomized phase 2 of adjuvant letrozole vs observation (GOG Foundation)
- **Effect size:** — — disease-free survival (primary)
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### Meric-Bernstam F/Lee JY (2024) — J Clin Oncol

**Reference:** [PMID 37870536](#) · **Type:** trial · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd)
- **Indication / model:** HER2-expressing (IHC 3+/2+) advanced solid tumors after prior therapy (DESTINY-PanTumor02) (2L+); HER2 IHC 3+ or 2+ — never stained in this patient; tumor-agnostic approval is IHC 3+ only
- **Design:** multicohort single-arm phase 2 basket
- **n:** 267
- **Effect size:** ORR 0.371 — ORR 37.1% overall; 61.3% in IHC 3+
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

## D'Angelo SP/Van Tine BA (2024) — Lancet

**Reference:** PMID 38554725 · **Type:** trial · **Inclusion:** included

- **Intervention:** afamitresgene autoleucel
- **Indication / model:** advanced synovial sarcoma / myxoid round cell liposarcoma (SPEARHEAD-1) (2L+); HLA-A\*02 allele + MAGE-A4 expression  $\geq 1+$  in  $\geq 10\%$  of cells (central IHC); both unassessable in this patient (HLA untyped, antigen unstained)
- **Design:** single-arm phase 2 of afamitresgene autoleucel (MAGE-A4 TCR-T)
- **n:** 52
- **Effect size:** ORR 0.37 — ORR 37% (synovial sarcoma 39%)
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

## —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** MGC026
- **Indication / model:** advanced solid tumors including sarcoma (named cohort) (2L+); B7-H3-directed; no expression gate at entry (97% of soft-tissue sarcomas B7-H3-positive, PMID 39478506)
- **Design:** phase 1 dose escalation/expansion of MGC026 (B7-H3 ADC)
- **Effect size:** — — safety/RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** GSK5764227 (HS-20093)
- **Indication / model:** advanced/metastatic solid tumors after progression on available standard therapy (EMBOLD) (2L+); B7-H3-directed; no expression gate at entry
- **Design:** phase 1 dose escalation/expansion of GSK5764227 (HS-20093, B7-H3 ADC) alone and in combinations
- **Effect size:** — — safety/RP2D
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Ingham/Schwartz (2023) — J Clin Oncol

**Reference:**PMID 37467452 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Olaparib + temozolomide (NCI Protocol 10250)
- **Indication / model:** advanced uterine leiomyosarcoma after  $\geq 1$  prior line (2L+); 22 evaluable; median age 55; 59% had  $\geq 3$  prior lines; no HRD entry gate — all comers with paired biopsies (WES/RNAseq + RAD51 foci assay)
- **Design:** single-arm open-label multicentre phase 2, one-stage binomial design ( $\geq 5/22$  responses = promising)
- **n:** 22
- **Effect size:** 27 % — ORR (overall)
- **Toxicities:** neutropenia  $G \geq 3$  (n=22, 75%); thrombocytopenia  $G \geq 3$  (7/22, 32%); anemia  $G \geq 3$  (7/22, 32%); discontinuation for toxicity (3/22, 14%); olaparib dose reduction (8/22, 36%)
- **Case fit:** partial
- **Notes:** The strongest non-cytotoxic signal in uLMS and the branch her pending HRD/RAD51-relevant profiling gates. n=22 and single-arm: the 27% ORR and the 11.2 vs 5.4-month RAD51 split are hypothesis-grade until the randomized successor (NCT05432791) reads out. An erratum (J Clin Oncol 2023;41:5205) was issued. AE and subgroup detail from PMC full text (PMC10852403).

## Dall/Barker (2023) — J Exp Clin Cancer Res

**Reference:**PMID 37143137 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibition in HR-deficient uterine leiomyosarcoma (PDX)
- **Indication / model:** patient-derived xenografts from HRD uterine leiomyosarcoma, including a paired PDX derived after PARP-inhibitor exposure; WGS/WES/panel screening of a 58-patient uLMS cohort
- **Design:** Homozygous BRCA2 deletion and COSMIC signature 3 with high genome-wide LOH define an HR-deficient subset; PARP inhibition converts unrepaired single-strand damage into lethal double-strand breaks, and PARP1-selective trapping does so with less of the off-target PARP2 burden.
- **n:** 58 patients screened; 5 (9%) HRD; PDX generated from HRD cases including one post-PARPi pair
- **Effect size:** strong — All 13 uLMS analysed by WGS carried a dominant signature 3, and 5 of 58 patients had HRD uLMS, in every confirmed case through homozygous BRCA2 deletion. PDX responses were fastest, deepest and most durable with the PARP1-selective AZD5305 rather than olaparib or olaparib plus cisplatin, holding even in a PDX derived after clinical PARPi exposure that had acquired PRKDC resistance mutations.
- **Variance:** vs olaparib alone and olaparib plus cisplatin compared against the PARP1-selective AZD5305; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Small PDX numbers and a paper that interleaves patient outcomes with model pharmacology. The agent that worked best, AZD5305, is not the one this patient could reach; the accessible branch is olaparib plus temozolomide.

## Brignole/Ponzoni (2023) — J Immunother Cancer

**Reference:**PMID 37775116 · **Type:** preclinical · **Inclusion:** included



- **Intervention:** Vobramitamab duocarmazine (MGC018) with bystander killing
- **Indication / model:** human neuroblastoma cell lines, multicellular tumour spheroids, and orthotopic, pseudometastatic and resected mouse neuroblastoma models; co-culture of B7-H3-negative murine NX-S2 cells with B7-H3-positive human cells
- **Design:** A cleavable valine-citrulline linker releases membrane-permeable duocarmycin, so payload liberated inside an antigen-positive cell can cross into neighbouring antigen-negative cells — the bystander effect that makes heterogeneous or low antigen tractable.
- **Effect size:** strong — Cytotoxicity ran from IC50 5.1-53.9 ng/mL in monolayer and 17.8-364 ng/mL in spheroids. B7-H3-negative murine cells, untouched by the ADC alone, were killed when co-cultured with B7-H3-positive cells, and treatment delayed or arrested relapse across orthotopic and resected models without weight loss or haematologic toxicity.
- **Variance:** vs irrelevant (anti-CD20) duocarmycin ADC; topotecan-temozolomide as the standard-of-care comparator; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Neuroblastoma throughout, with no sarcoma model and no leiomyosarcoma. The bystander demonstration is the part that transfers, and it is the mechanistic argument for a B7-H3 ADC where expression is heterogeneous rather than uniform.

## Crane/Federman (2023) — Sarcoma

**Reference:** [PMID 37333052](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** FAP expression in sarcoma: stroma and tumour cells
- **Indication / model:** IHC on 63 bone and soft-tissue tumour samples with 30 adjacent normal tissues, scored separately for stromal and tumour-cell compartments; RNA-seq comparison across 10,626 samples and survival analysis in 168 sarcomas
- **Design:** FAP is a marker of activated fibroblasts. In mesenchymal tumours the malignant cell is itself of fibroblastic lineage, so the antigen is not confined to the stromal compartment the way it is in carcinomas.
- **n:** n=63 tumours; n=30 adjacent normal; n=168 sarcomas for survival
- **Effect size:** Intensity  $\geq 2$  with density  $\geq 25\%$  appeared in stromal cells of 77.7% of tumours and in tumour cells of 50.7%. Sarcomas ranked among the highest FAP-expressing cancer types by RNA-seq, and FAP level carried no overall-survival association.
- **Variance:** vs adjacent normal tissue and positive controls; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Mixed bone and soft-tissue histologies with no leiomyosarcoma breakout, and no functional or therapeutic arm. It does change the mechanistic argument: in sarcoma a FAP radioligand may deliver dose directly to tumour cells rather than relying solely on crossfire from stroma.

## Galis/Rajtik (2023) — Cardiovasc Toxicol

**Reference:** [PMID 37119387](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Hypoxic preconditioning and doxorubicin cardiomyopathy
- **Indication / model:** Wistar rats given doxorubicin, with or without intermittent hypoxic preconditioning beforehand; no tumour present
- **Design:** Preconditioning raised SERCA2a expression, which the authors link to preserved calcium handling;

HIF-1 $\alpha$  levels did not change, and TRPV1/TRPV4 loss and suppressed CaMKII activation followed doxorubicin regardless.

- **Effect size:** weak — Preconditioned rats retained more cardiac output than rats given doxorubicin alone, with higher SERCA2a on western blot and no change in HIF-1 $\alpha$ .
- **Variance:** vs doxorubicin without preconditioning; untreated rats; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Cycled normobaric hypoxia in healthy rats is not continuous residence at 8,500 ft, no tumour was present, and the mechanism is inferred from protein levels rather than measured function. Logged to record that the altitude question has no translatable preclinical answer — direction of effect is unresolved and should not be used either to reassure or to alarm.

### —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** lurbinectedin + doxorubicin
- **Indication / model:** metastatic leiomyosarcoma, first line (SaLuDo) (1L); none required; anthracycline-naive with ECOG  $\leq 1$  and LVEF  $> 50\%$  required
- **Design:** randomized phase 3: lurbinectedin + doxorubicin vs doxorubicin alone
- **Effect size:** — — PFS (primary)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** olaparib + temozolomide
- **Indication / model:** advanced uterine leiomyosarcoma after  $\geq 2$  prior lines including an anthracycline (3L+); no HRD gate; prior PARP inhibitor, temozolomide or dacarbazine excluded
- **Design:** randomized phase 2/3: olaparib + temozolomide vs investigator's choice (trabectedin or pazopanib)
- **Effect size:** — — PFS (primary)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** HS-20093
- **Indication / model:** relapsed/refractory osteosarcoma and other sarcomas after first-line systemic treatment (ARTEMIS-002) (2L+); B7-H3-directed; no expression gate; prior B7-H3 therapy excluded
- **Design:** phase 2 of HS-20093 (B7-H3 ADC) in sarcoma
- **Effect size:** — — ORR (primary)
- **Case fit:** weak

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** vobramitamab duocarmazine
- **Indication / model:** metastatic castration-resistant prostate cancer and other solid tumors (TAMARACK program context) (2L+); B7-H3-directed
- **Design:** phase 2 of vobramitamab duocarmazine (B7-H3 ADC, duocarmycin payload); terminated
- **Effect size:** — — ORR/rPFS (primary)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** PARP inhibitor + SBRT + immune checkpoint blockade
- **Indication / model:** metastatic or advanced high-grade bone/soft-tissue sarcoma after ≥1 systemic regimen (PRIMA) (2L+); no PARP-inhibitor pretreatment; SBRT-amenable lesions required
- **Design:** phase 2 of PARP inhibition + stereotactic body radiotherapy + PD-1 blockade
- **Effect size:** — — ORR/PFS (primary)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Pautier/Duffaud (2022) — Lancet Oncol

**Reference:** [PMID 35835135](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + trabectedin, then trabectedin maintenance (LMS-04)
- **Indication / model:** metastatic or unresectable uterine or soft-tissue leiomyosarcoma, chemotherapy-naïve (1L); ECOG 0-1 required; 67 of 150 patients (45%) had uterine LMS; randomization stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles in both arms
- **Design:** randomized open-label phase 3 (French Sarcoma Group, 20 centres), BICR-assessed primary endpoint
- **n:** 150
- **Effect size:** 12.2 months — median PFS (BICR)
- **Variance:** 95% CI 10.1–15.6;  $p < 0.0001$
- **Toxicities:** neutropenia  $G \geq 3$  (59/74, 80%); thrombocytopenia  $G \geq 3$  (35/74, 47%); anemia  $G \geq 3$  (23/74, 31%); febrile neutropenia  $G \geq 3$  (21/74, 28%); serious adverse event (15/74, 20%); treatment-related death (cardiac failure) G5 (0/74, 0%)
- **Case fit:** strong
- **Notes:** Primary report of the regimen she is about to start; pivotal for the LMS-04 strategy. The PMID a search commonly surfaces, 36174625, is a letter about this trial, not the trial. Per-term G3-4 rates above are from the combination phase; maintenance-phase safety detail sits behind the paywall (no PMC; CT.gov NCT02997358)

has no posted results).

### Sütcüoğlu/— (2022) — Lancet Oncol

**Reference:**PMID 36174625 · **Type:** clinical · **Inclusion:** excluded — Letter commenting on the LMS-04 trial, not the trial report. Logged because this PMID commonly surfaces in searches for the regimen; the primary publication is PMID 35835135.

- **Intervention:** Doxorubicin + trabectedin (LMS-04) — correspondence
- **Notes:** Excluded: Letter commenting on the LMS-04 trial, not the trial report. Logged because this PMID commonly surfaces in searches for the regimen; the primary publication is PMID 35835135.

### Li/Duan (2022) — Biomed Pharmacother

**Reference:**PMID 35367753 · **Type:** preclinical · **Inclusion:** excluded — Synovial sarcoma, where CDK7 dependency is argued from SS18-SSX-driven transcriptional addiction — a mechanism this tumour does not share. In vitro only (siRNA and BS-181), with no RB1 stratification.

- **Intervention:** CDK7 inhibition in synovial sarcoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Synovial sarcoma, where CDK7 dependency is argued from SS18-SSX-driven transcriptional addiction — a mechanism this tumour does not share. In vitro only (siRNA and BS-181), with no RB1 stratification.

### Zboralski/Osterkamp (2022) — Eur J Nucl Med Mol Imaging

**Reference:**PMID 35608703 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** FAP-2286 radioligand — preclinical imaging and therapy
- **Indication / model:** FAP-expressing HEK293 tumours and sarcoma patient-derived xenografts in mice; IHC and autoradiography on primary human cancer specimens including sarcoma
- **Design:** A cyclic peptide binder coupled to a chelator: it holds at FAP-positive tissue long enough for a bound beta emitter to deposit dose, with renal clearance keeping normal-tissue exposure low.
- **Effect size:** moderate — Tumour uptake was rapid and persistent across all three radionuclides with minimal normal-tissue signal, and <sup>177</sup>Lu-FAP-2286 produced antitumour activity in FAP-expressing HEK293 tumours and in sarcoma PDX without significant weight loss. Tumour retention and suppression outlasted FAPI-46.
- **Variance:** vs FAPI-46, a small-molecule FAP agent, as the head-to-head comparator; non-FAP-expressing tumours for specificity; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The sarcoma evidence is PDX efficacy without histology-level detail, and PDX stroma is host mouse tissue, so what the ligand engaged in those tumours is not resolvable from the report. The engineered HEK293 model tests direct antigen binding rather than the stromal-crossfire premise.

### Belgiovine/D'Incalci (2021) — Eur J Immunol

**Reference:**PMID 34570376 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin before anti-PD-1 in a checkpoint-resistant model
- **Indication / model:** murine fibrosarcoma poorly responsive to PD-1 blockade, in immunocompetent mice
- **Design:** Removing tumour-associated macrophages lifts myeloid suppression; T-cell and interferon-response transcription rises, Pdc1 goes up, and the tumour becomes answerable to PD-1 blockade.
- **Effect size:** moderate — Treated tumours upregulated CD3, CD8, perforin, granzyme B and IFN-response genes, with a marked rise in Pdc1. Trabectedin given ahead of anti-PD-1 improved efficacy in a model that otherwise resists checkpoint blockade.
- **Variance:** vs anti-PD-1 alone, trabectedin alone, and untreated tumours; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Mouse fibrosarcoma, not LMS, and the benefit depended on trabectedin coming first. No preclinical data exist for adding checkpoint blockade to an LMS-04-style doxorubicin-trabectedin schedule.

## —/— (2021)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** [177Lu]Lu-FAP-2286
- **Indication / model:** advanced solid tumors with FAP-2286 uptake (LuMIERE) (2L+); tumor uptake on [68Ga]Ga-FAP-2286 PET required — FAP status unmeasured in this patient
- **Design:** phase 1/2 theranostic: [68Ga]Ga-FAP-2286 PET selection, then [177Lu]Lu-FAP-2286 therapy, monotherapy and chemo combinations
- **Effect size:** — — safety/RP2D (phase 1); ORR (phase 2)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Baumann/Chen (2020) — J Surg Oncol

**Reference:** [PMID 32588468](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** SBRT for sarcoma pulmonary metastases (MGH/Penn)
- **Indication / model:** oligometastatic sarcoma lung metastases (any); 44 patients / 56 lesions; 82% prior chemotherapy, 66% prior pulmonary resection; median lesion 2.0 cm
- **Design:** retrospective two-institution cohort
- **n:** 44
- **Effect size:** 90 % — local control at 24 months
- **Variance:** 95% CI 84–96
- **Toxicities:** chest-wall toxicity G1-2 (3/44, 6.8%); pneumonitis G1-2 (1/44, 2.3%); any adverse event G $\geq$ 3 (0/44, 0%)
- **Case fit:** partial
- **Notes:** Largest multi-institution SBRT series in sarcoma lung metastases: excellent lesion-level control, low toxicity, but OS tracks systemic disease. Local control is not survival, and candidacy assumes discrete targetable lesions — her infiltrative pattern and 8,500-ft pulmonary baseline are both open questions for this modality.

## Tap/Jones (2020) — JAMA

**Reference:**PMID 32259228 · **Type:** clinical · **Inclusion:** excluded — Olaratumab targets PDGFR-alpha, which is not one of her nominated targetable features, and the drug's STS indication was withdrawn after this negative confirmatory trial. Reviewed for its doxorubicin-control benchmark: median OS 21.9 months in the LMS stratum with modern supportive care.

- **Intervention:** Doxorubicin + olaratumab (ANNOUNCE)
- **Notes:** Excluded: Olaratumab targets PDGFR-alpha, which is not one of her nominated targetable features, and the drug's STS indication was withdrawn after this negative confirmatory trial. Reviewed for its doxorubicin-control benchmark: median OS 21.9 months in the LMS stratum with modern supportive care. — Kept out of synthesis per the mechanism-scope rule; the control-arm OS is context for interpreting single-agent doxorubicin expectations, nothing more.

## Zhang/Wong (2020) — Cancer Cell

**Reference:**PMID 31883968 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Selective CDK7 inhibition (YKL-5-124) in RB1/TP53-null tumours
- **Indication / model:** human small-cell lung cancer lines and multiple aggressive murine SCLC models; SCLC is near-universally RB1- and TP53-null. YKL-5-124 given with and without anti-PD-1
- **Design:** CDK7 sits at the junction of cell-cycle CDK activation and RNA Pol II transcription. Inhibiting it in cells that have already lost the RB1 restriction point disrupts cycle progression, forces replication stress and genome instability, and the resulting cytosolic DNA drives immune signalling.
- **Effect size:** strong — CDK7 inhibition blocked cycle progression and induced replication stress and genome instability while switching on immune-response signalling. The consequent T-cell surveillance was amplified by checkpoint blockade, and YKL-5-124 plus anti-PD-1 extended survival across several aggressive SCLC models.
- **Variance:** vs vehicle; anti-PD-1 alone; CDK7 inhibitor alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** SCLC is RB1-null like the trial's target population but is a different lineage entirely, and YKL-5-124 is not REC-617. No CDK7 data of any kind exist in leiomyosarcoma, which makes the RB1-null CDK7 door the thinnest mechanistic branch in this case.

## Goncalves/Tomita (2020) — ACS Pharmacol Transl Sci

**Reference:**PMID 33344901 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibitors in sarcoma lines: telomere length, not ALT, tracked sensitivity
- **Indication / model:** 17 osteosarcoma-derived cell lines characterised for telomere maintenance mechanism and telomere length; AZD-6738 (ceralasertib), VE-822 and BAY-1895344
- **Design:** ATR sustains chromosome stability when telomeres are critically short; the dependency follows telomere length rather than the mechanism used to maintain it.
- **n:** n=17 cell lines (8 ALT-positive)
- **Effect size:** negative — Short telomeres, not ALT, predicted hypersensitivity to all three ATR inhibitors, with chromosome bridges and cell death inside 24 hours. ALT-positive lines carried excessive telomere length and were no more sensitive than telomerase-positive controls.
- **Variance:** vs ALT-positive and telomerase-positive long-telomere lines compared against short-telomere lines; translatability: med
- **Toxicities:** n/a (preclinical)

- **Case fit:** cross\_tumor\_only
- **Notes:** Osteosarcoma rather than LMS, and in-vitro only. Within sarcoma this is the most directly damaging evidence against treating ALT positivity as an ATR-inhibitor biomarker, and it points to telomere length as the measurement that would actually matter.

## Petitprez/Fridman (2020) — Nature

**Reference:**PMID 31942077 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Sarcoma immune classes and B-cell-rich tertiary lymphoid structures
- **Indication / model:** gene-expression profiling of 608 soft-tissue sarcoma tumours with in-situ validation in an independent cohort and a phase 2 pembrolizumab trial
- **Design:** Response to PD-1 blockade in sarcoma tracks with the composition of the microenvironment rather than with tumour mutational burden. B-cell-rich tertiary lymphoid structures mark the class where antigen presentation is organised enough for checkpoint blockade to act on.
- **n:** n=608 tumours for classification
- **Effect size:** moderate — Five microenvironment phenotypes emerged: two immune-low, two immune-high, one highly vascularised. B-cell content was the strongest prognostic factor even after accounting for CD8 T cells and cytotoxic signature, and the TLS-rich class E showed both better survival and a high response rate to pembrolizumab.
- **Variance:** vs not applicable — classification study with a prospective response cohort attached; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Leiomyosarcoma falls predominantly into the immune-low and highly vascularised classes, which is the structural reason checkpoint blockade underperforms in this histology. Response rates are reported by class, not by histology, so no LMS-specific number can be read off this paper.

## Marabelle A/Bang YJ (2020) — Lancet Oncol

**Reference:**PMID 32919526 · **Type:** trial · **Inclusion:** included

- **Intervention:** pembrolizumab
- **Indication / model:** advanced solid tumors, TMB-high and MSI-H cohorts (KEYNOTE-158) (2L+); TMB ≥10 mut/Mb (FoundationOne CDx) or MSI-H/dMMR — both unmeasured in this patient
- **Design:** multicohort single-arm phase 2 basket
- **n:** 790
- **Effect size:** ORR 0.29 — ORR 29% (TMB-high) vs 6% (non-TMB-high)
- **Case fit:** partial
- **Notes:** (toxicity table not extracted)

## Hong DS/Drilon A (2020) — Lancet Oncol

**Reference:**PMID 32105622 · **Type:** trial · **Inclusion:** included

- **Intervention:** larotrectinib
- **Indication / model:** TRK fusion-positive solid tumors, any histology (pooled NAVIGATE/SCOUT/phase 1) (2L+); NTRK1/2/3 in-frame fusion — requires the RNA-based fusion assay she has not had



- **Design:** pooled analysis of three single-arm phase 1/2 trials
- **n:** 159
- **Effect size:** ORR 0.79 — ORR 79%
- **Case fit:** partial
- **Notes:** (toxicity table not extracted)

## —/— (2020)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** pulmonary suffusion + metastasectomy
- **Indication / model:** sarcoma or colorectal metastases to lung undergoing pulmonary metastasectomy (n/a); none; pulmonary function sufficient for planned metastasectomy required
- **Design:** phase 1/2 of pulmonary suffusion (regional lung chemotherapy) at the time of metastasectomy, with SBRT/ablation hybrid approaches permitted
- **Effect size:** — — feasibility/safety; control of minimal residual disease
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Blay/Maki (2019) — Br J Cancer

**Reference:** [PMID 31065111](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eribulin vs dacarbazine — leiomyosarcoma subgroup of Study 309
- **Indication / model:** advanced leiomyosarcoma (42% uterine) after  $\geq 2$  prior regimens including an anthracycline (2L+); 309 LMS patients (eribulin 157, dacarbazine 152); 42% uterine primary; ECOG  $\leq 2$
- **Design:** preplanned subgroup analysis of the randomized phase 3 Study 309
- **n:** 309
- **Effect size:** 0.93 HR — median OS (LMS subset)
- **Variance:** 95% CI 0.71–1.2;  $p=0.57$
- **Toxicities:** any treatment-emergent AE  $G \geq 3$  (n=157, 69%)
- **Case fit:** partial
- **Notes:** The honest read for her histology: eribulin matched but did not beat dacarbazine in LMS on every endpoint. Keeps eribulin as a tolerable later-line option rather than a preferred one; subgroup of a trial powered for the whole L-sarcoma population.

## Böhm/Barth (2019) — Sarcoma

**Reference:** [PMID 30804704](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Palbociclib in Rb-positive vs Rb-negative LMS cell lines
- **Indication / model:** SK-LMS-1 (Rb-expressing) and SK-UT-1 (Rb-negative) leiomyosarcoma cell lines; genomic profiling of 18 LMS samples; p16/CDK4/CDK6/p-Rb IHC across 159 samples from 99 LMS patients
- **Design:** Palbociclib blocks CDK4/6-mediated Rb phosphorylation. With no Rb protein present there is nothing downstream to hold, and the drug has no substrate through which to arrest the cycle.
- **n:** 18 tumour samples across 9 patients for profiling; 99 patients / 159 samples for IHC
- **Effect size:** moderate — Palbociclib collapsed phospho-Rb (Ser780), slowed proliferation and forced G0/G1



arrest in Rb-positive SK-LMS-1, while Rb-negative SK-UT-1 was unaffected. CDK4 amplification turned up in 5 of 18 LMS samples, and an IHC-defined potential-responder phenotype in 29 of 99 patients.

- **Variance:** vs untreated cells; the Rb-negative line serves as the internal biological control; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Two cell lines, no in-vivo arm, and the responder phenotype is an IHC construct never tested prospectively. It is nonetheless the cleanest LMS-specific demonstration that one Rb stain sorts responders from non-responders.

## Majzner/Mackall (2019) — Clin Cancer Res

**Reference:** [PMID 30655315](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** B7-H3 CAR T cells and the antigen-density threshold
- **Indication / model:** xenograft models of osteosarcoma, medulloblastoma and Ewing sarcoma; B7-H3 CAR T cells tested against targets differing in surface antigen density
- **Design:** CAR T-cell killing scales with the number of antigen molecules per target cell. Below a threshold density the synapse fails to trigger, which is what creates a therapeutic window despite low-level B7-H3 on normal tissue.
- **Effect size:** strong — B7-H3 CAR T cells caused regression of established solid tumours across several xenograft models. Efficacy depended heavily on surface antigen density and fell away sharply against low-expressing targets.
- **Variance:** vs control T cells without the B7-H3 CAR; target cells expressing low versus high surface B7-H3; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Pediatric solid-tumour xenografts, no leiomyosarcoma model and no ADC. The density dependence is the transferable point: it is why an IHC-positive result alone does not establish that a B7-H3 agent will work, and why the 69% 'high' stratum matters more than the 97% positivity figure.

## —/— (2019)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 (PRAME TCR-T)
- **Indication / model:** recurrent/refractory PRAME-positive solid tumors, HLA-A02:01-restricted (*ACTengine* IMA203) (2L+); HLA-A02:01 genotype + PRAME expression by sponsor qPCR — both unmeasured; ECOG 0-1 required
- **Design:** phase 1/2 basket of IMA203 / IMA203CD8 PRAME TCR-T, monotherapy or with nivolumab
- **Effect size:** — — safety/RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Chudasama/Fröhling (2018) — Nat Commun

**Reference:** [PMID 29321523](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** LMS genomic landscape: TP53/RB1 loss, ALT, and BRCAness
- **Indication / model:** whole-exome and transcriptome sequencing of leiomyosarcoma tumours plus short-term cultured LMS cells treated with olaparib or cisplatin
- **Design:** Near-universal TP53 and RB1 inactivation with widespread copy-number change, chromothripsis and whole-genome duplication; telomere maintenance runs through ALT, and structural-rearrangement burden with the associated mutational signatures produces a BRCAness state that PARP inhibition can act on.
- **Effect size:** moderate — TP53 and RB1 were inactivated in nearly every tumour, alternative telomere lengthening was present in 78% of cases, and recurrent alterations hit ATRX, RBL2 and SP100. Most tumours carried hallmarks of BRCAness, and cultured LMS cells were sensitive to olaparib and to cisplatin.
- **Variance:** vs untreated cultured LMS cells for the drug-sensitivity work; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mixed LMS sites rather than uterine-only, and the drug sensitivity is short-term culture with no in-vivo arm. The near-universal RB1 loss is the single most decision-relevant number here: it shifts the prior for an Rb IHC toward null and away from the abemaciclib door.

## Castroviejo-Bermejo/Serra (2018) — EMBO Mol Med

**Reference:**PMID 30377213 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** RAD51 nuclear-foci assay as a functional HRD readout
- **Indication / model:** breast cancer patient-derived xenograft panels treated with olaparib, plus archival clinical tumour samples including PALB2-related cancers
- **Design:** RAD51 nuclear foci in replicating cells report whether homologous recombination is actually executing at the time of testing, which a sequencing panel cannot show — a gene may be intact and the pathway still broken, or mutated and functionally reverted.
- **Effect size:** strong — The RAD51 score separated PARPi-sensitive from PARPi-resistant PDX and outperformed the genomic HRD score in an independent panel. In archival samples the assay classified every PALB2-related tumour as HR-deficient, extending the deficient population beyond BRCA1/2 carriers.
- **Variance:** vs genomic HRD score and exome/BRCA1-methylation analysis run head-to-head against the RAD51 score; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Breast PDX and breast archival tissue; the assay needs proliferating cells and geminin co-staining, so specimen quality governs interpretability. Its only uLMS outcome data are the 22-patient NCI 10250 trial, where RAD51-defined HR deficiency separated PFS of 11.2 from 5.4 months.

## Seddon/Beare (2017) — Lancet Oncol

**Reference:**PMID 28882536 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel vs doxorubicin (GeDDiS)
- **Indication / model:** previously untreated locally advanced or metastatic soft-tissue sarcoma (1L); LMS-enriched: 118/257 (46%) leiomyosarcoma, of whom 71 (28% of trial) uterine LMS; Trojani grade 2-3
- **Design:** randomized controlled phase 3 (GeDDiS, ISRCTN07742377), 24-week PFS-rate primary endpoint
- **n:** 257
- **Effect size:** 1.28 HR — alive and progression-free at 24 weeks
- **Variance:** 95% CI 0.99–1.65; p=0.06

- **Toxicities:** neutropenia G $\geq$ 3 (25/126, 20%); febrile neutropenia G $\geq$ 3 (15/126, 12%); fatigue G $\geq$ 3 (17/126, 14%); oral mucositis G $\geq$ 3 (2/126, 2%); pain G $\geq$ 3 (13/126, 10%)
- **Case fit:** partial
- **Notes:** The head-to-head that keeps anthracycline first: gem-doce showed no superiority even in an LMS-enriched population, including the uterine subset (interaction p=0.38). Its practical role for her is first-line fallback if baseline LVEF or pulmonary reserve at 8,500 ft rules out doxorubicin.

## Hensley/Demetri (2017) — Gynecol Oncol

**Reference:**PMID 28651804 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Trabectedin vs dacarbazine — uterine LMS subgroup of SAR-3007
- **Indication / model:** advanced uterine leiomyosarcoma after anthracycline-based chemotherapy (2L+); 232 of 577 randomized patients had uLMS (trabectedin 144, dacarbazine 88) — the largest uLMS randomized dataset in this line
- **Design:** post hoc subgroup analysis of the randomized phase 3 SAR-3007 trial
- **n:** 232
- **Effect size:** 0.57 HR — median PFS (uLMS subset)
- **Variance:** 95% CI 0.41–0.81; p=0.0012
- **Toxicities:** neutropenia G3 (29/140, 21%); leukopenia G3 (29/140, 21%); anemia G3 (27/140, 19%); AST/ALT elevation G $\geq$ 3 (n=140, 33%); thrombocytopenia G3 (7/140, 5%); treatment-related death G5 (2/140, 1.4%)
- **Case fit:** partial
- **Notes:** The uLMS-specific randomized evidence for trabectedin after anthracycline: PFS benefit holds (HR 0.57), OS does not. Post hoc subgroup of a trial not powered for it — treat the point estimates accordingly. AE data from PMC full text (PMC5783302).

## Seagle/Shahabi (2017) — Gynecol Oncol

**Reference:**PMID 28317559 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Prognosis and treatment of uterine leiomyosarcoma (NCDB)
- **Indication / model:** uterine leiomyosarcoma, all stages (metastatic subset analyzed for chemotherapy effect) (any); 7,455 women, US National Cancer Database 1998-2013; matched-cohort analyses for surgery/chemotherapy questions
- **Design:** retrospective national registry cohort with matched analyses
- **n:** 7455
- **Effect size:** 19.4 months — median OS with vs without chemotherapy, metastatic uLMS
- **Variance:** 95% CI 16.4–23.0; p=<0.001
- **Toxicities:** Registry analysis; no treatment-safety data.
- **Case fit:** partial
- **Notes:** Population-level anchor for metastatic-uLMS prognosis in the pre-LMS-04 era: median OS around 10-19 months depending on chemotherapy receipt, against which LMS-04's 33-month combination-arm median reads as a different disease course. Registry data with confounding by indication; use for expectation-setting, not treatment choice.

## Demetri/— (2017) — J Clin Oncol

**Reference:**PMID 28854066 · **Type:** clinical · **Inclusion:** excluded — Liposarcoma subgroup analysis of eribulin

Study 309 — wrong histology for this uLMS case; it is the basis of the liposarcoma-only FDA label. Her histology is covered by the LMS subgroup row (PMID 31065111).

- **Intervention:** Eribulin — liposarcoma subgroup of Study 309
- **Notes:** Excluded: Liposarcoma subgroup analysis of eribulin Study 309 — wrong histology for this uLMS case; it is the basis of the liposarcoma-only FDA label. Her histology is covered by the LMS subgroup row (PMID 31065111).

## Belgiovine/D'Incalci (2017) — Br J Cancer

**Reference:**PMID 28683469 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Lurbinectedin — macrophage and monocyte depletion
- **Indication / model:** tumour-bearing mice including a lurbinectedin-resistant tumour variant; purified human monocytes in vitro
- **Design:** Same structural class as trabectedin: transcription and repair inhibition in tumour cells, plus caspase-8-dependent apoptosis of monocytes and suppression of CCL2, CXCL8 and VEGF at sub-lethal concentrations.
- **Effect size:** moderate — Circulating monocytes, intratumoural macrophages and vessel density all fell, including in the resistant variant where the tumour cells themselves were untouched. Below the apoptotic dose, monocyte adhesion and migration collapsed alongside downregulation of Rho-GTPase family genes.
- **Variance:** vs untreated tumour-bearing mice; resistant variant used to separate tumour-cell kill from microenvironment effects; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** No sarcoma model and no doxorubicin combination, which is what the LMS trial (NCT06088290) actually tests. Accrual on that trial closed before this patient's diagnosis was corrected, so the row is mechanistic context.

## Pignochino/Grignani (2017) — Mol Cancer

**Reference:**PMID 28454547 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin + olaparib (PARP1-dependent synergy)
- **Indication / model:** multiple sarcoma histotypes in vitro and subcutaneous plus orthotopic mouse xenografts, including the DMR leiomyosarcoma model and SJSA-1 osteosarcoma
- **Design:** Trabectedin-induced single- and double-strand breaks activate PARP1; blocking PARP1 leaves that damage unresolved, driving G2/M arrest and apoptosis. The size of the effect scales with how much PARP1 the tumour expresses.
- **Effect size:** moderate — Adding olaparib deepened DNA damage, G2/M arrest and apoptosis over trabectedin alone, with antitumour and antimetastatic activity in mice. Synergy was strong in the DMR leiomyosarcoma model and absent in SJSA-1 osteosarcoma, and it moved up or down when PARP1 was overexpressed or silenced.
- **Variance:** vs trabectedin alone, PARP inhibitor alone, vehicle; PARP1 overexpression and silencing as functional controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The responsive LMS line is not uterine, and PARP1 expression, the proposed predictor, has never been

measured in this patient. No clinical test of this combination in LMS exists.

## Francis/Hunt (2017) — Mol Cancer Ther

Reference: [PMID 28619757](#) · Type: preclinical · Inclusion: included

- **Intervention:** Palbociclib priming of Rb-positive sarcoma, including LMS PDX
- **Indication / model:** SK-LMS1 leiomyosarcoma and HT-1080 fibrosarcoma lines; SK-LMS1 xenografts and Rb-positive patient-derived xenografts from leiomyosarcoma patients
- **Design:** Palbociclib holds Rb-positive cells in a reversible G1 arrest; on release they move through the cycle within 6-24 hours, and a synchronised S-G2 population is more vulnerable to agents that act there.
- **Effect size:** strong — Palbociclib arrest was fully reversible, and priming Rb-positive sarcoma cells this way made them more sensitive to doxorubicin and to Wee1 inhibition. The palbociclib-AZD1775 synergy held in SK-LMS1 xenografts and in Rb-positive leiomyosarcoma PDX.
- **Variance:** vs single-agent palbociclib, single-agent AZD1775, vehicle; concurrent versus sequential dosing compared; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** LMS PDX but no stated uterine primary. The sensitisation depends on sequencing — arrest, release, then the S-G2 agent — which is the opposite of concurrent dosing (see McClendon 2012).

## —/— (2017)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** anlotinib (catequentinib)
- **Indication / model:** advanced alveolar soft part sarcoma, leiomyosarcoma, synovial sarcoma (APROMISS); LMS indications closed (2L+); none required
- **Design:** phase 3 of anlotinib (catequentinib) vs dacarbazine; LMS arms suspended, 'any sarcoma' indication E open
- **Effect size:** — — PFS (primary)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Demetri/Patel (2016) — J Clin Oncol

Reference: [PMID 26371143](#) · Type: clinical · Inclusion: included

- **Intervention:** Trabectedin vs dacarbazine after anthracycline failure (ET743-SAR-3007)
- **Indication / model:** advanced liposarcoma or leiomyosarcoma after anthracycline plus at least one further regimen (2L+); L-sarcomas only; uLMS was 40% of enrolment (see companion subgroup row); ECOG 0-1
- **Design:** randomized open-label phase 3, 2:1 allocation, OS primary endpoint
- **n:** 518
- **Effect size:** 0.55 HR — median PFS
- **Variance:**  $p < 0.001$
- **Toxicities:** neutropenia  $G \geq 3$  (n=340, 37%); ALT elevation  $G \geq 3$  (n=340, 25%); thrombocytopenia  $G \geq 3$  (n=340, 17%); anemia  $G \geq 3$  (n=340, 14%); rhabdomyolysis  $G \geq 3$  (n=340, 1.2%); treatment-related death G5

(7/340, 2.1%); discontinuation for toxicity (n=340, 12.6%)

- **Case fit:** partial
- **Notes:** Basis of the US trabectedin approval. For her it doubles as the single-agent safety dossier for the drug she will carry through LMS-04 maintenance — the hepatotoxicity and rhabdomyolysis rows are the ones her team will monitor. AE data from PMC full text (PMC5070559).

## Benson/van der Graaf (2016) — Gynecol Oncol

**Reference:**PMID 27012429 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pazopanib in uterine sarcoma (EORTC 62043/62072 retrospective)
- **Indication / model:** advanced uterine sarcoma (88.6% uLMS) within two prospective EORTC trials (2L+); 44 uterine sarcoma patients (39 uLMS, 37 high grade); heavily pretreated — 61.3% had  $\geq 2$  prior chemotherapy lines
- **Design:** retrospective subgroup analysis pooling EORTC phase 2 (62043) and phase 3 (PALETTE/62072) pazopanib-treated patients
- **n:** 44
- **Effect size:** 11 % — ORR (partial responses)
- **Variance:** 95% CI 3.8–24.6
- **Toxicities:** Safety not re-reported for the uterine subset; pazopanib AE profile carried on the PALETTE row.
- **Case fit:** partial
- **Notes:** Confirms pazopanib activity extends to uLMS (all five responses were LMS), with mPFS of 3 months in a heavily pretreated group. Retrospective pooling within prospective trials, n=44 — a supportive signal, not an independent efficacy estimate.

## Schöffski/Patel (2016) — Lancet

**Reference:**PMID 26874885 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eribulin vs dacarbazine (Study 309)
- **Indication / model:** advanced liposarcoma or leiomyosarcoma after  $\geq 2$  prior regimens including an anthracycline (2L+); intermediate/high-grade L-sarcomas; OS benefit was driven by the liposarcoma stratum (see LMS-subgroup row and excluded liposarcoma-subgroup row)
- **Design:** randomized open-label phase 3, OS primary endpoint
- **n:** 452
- **Effect size:** 0.77 HR — median OS
- **Variance:** 95% CI 0.62–0.95; p=0.0169
- **Toxicities:** any adverse event G $\geq 3$  (152/226, 67%); death on treatment G5 (10/226, 4%)
- **Case fit:** partial
- **Notes:** Whole-trial OS win that FDA translated into a liposarcoma-only label because the benefit concentrated there; for her LMS the drug is off-label with a neutral subgroup (next row). Per-term G3 rates not in abstract; Lancet full text paywalled, no PMC.

## Mäkinen/Vahteristo (2016) — PLoS Genet

**Reference:**PMID 26891131 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATRX loss and ALT in uterine leiomyosarcoma (target biology)



- **Indication / model:** whole-exome sequencing of 19 uterine leiomyosarcomas; ATRX IHC in 44 uLMS; telomere-specific FISH for the ALT phenotype
- **Design:** ATRX loads H3.3 at telomeric chromatin; losing it destabilises telomere maintenance and permits recombination-based lengthening, which is the ALT phenotype that ATR inhibition was proposed to exploit.
- **n:** n=19 exomes; n=44 tumours for IHC
- **Effect size:** TP53 was mutated in 6/19, ATRX in 5/19 (every alteration nonsense or frameshift) and MED12 in 4/19. ATRX expression was clearly reduced in 23 of 44 tumours by IHC, and every ATRX-low tumour tested ALT-positive by telomere FISH.
- **Variance:** vs not applicable — descriptive human tumour series; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive human series with no therapeutic arm; the ATR-inhibitor suggestion in its discussion is an extrapolation from Flynn 2015, not a result. The tight ATRX-IHC-to-ALT concordance is what makes one stain a usable surrogate here.

## Deeg/Rippe (2016) — Front Oncol

**Reference:**PMID 27602331 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition in ALT cells — failed replication
- **Indication / model:** panel of ALT-positive and telomerase-positive human cell lines plus isogenic pairs in which ALT was active or suppressed; ATR inhibitor VE-821
- **Design:** Same premise as Flynn 2015, tested with isogenic controls that hold genetic background constant while switching ALT on or off.
- **Effect size:** negative — No general ATR-inhibitor sensitivity followed from ALT status. Sensitivity differences tracked with line-to-line background rather than with the telomere maintenance mechanism.
- **Variance:** vs isogenic ALT-suppressed counterparts — the control Flynn 2015 lacked; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Small line panel and a single ATR inhibitor, so it does not exclude a narrower ALT-related vulnerability. It does undercut any claim that ATRX loss alone predicts ATR-inhibitor benefit.

## Navarria/Scorsetti (2015) — Eur J Cancer

**Reference:**PMID 25686482 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SBRT for lung metastases from soft-tissue sarcoma (Humanitas)
- **Indication / model:** lung metastases from soft-tissue sarcoma, unsuitable for resection,  $\leq 4$  lesions, controlled primary (any); 28 patients / 51 lesions; ECOG 1-2; LMS the commonest histology (36%); median interval from diagnosis to lung metastases 51 months
- **Design:** retrospective single-institution cohort
- **n:** 28
- **Effect size:** 96 % — actuarial 5-year local control
- **Toxicities:** any adverse event  $G \geq 3$  (0/28, 0%)
- **Case fit:** partial
- **Notes:** Small, highly selected ( $\leq 4$  lesions, controlled primary, long median disease interval), which explains the striking survival figures; that selection is the caveat to carry, not the local-control number. Supports SBRT as the non-surgical local option if induction converts her disease to oligometastatic anatomy.



## Flynn/Zou (2015) — Science

Reference: [PMID 25593184](#) · Type: preclinical · Inclusion: included

- **Intervention:** ATR inhibition in ALT-positive cells (original claim)
- **Indication / model:** ALT-positive and telomerase-positive human cancer cell lines, including ATRX-manipulated pairs; ATR inhibitors in vitro
- **Design:** ATRX loss uncouples cell-cycle control of the telomeric transcript TERRA, leaving RPA bound to telomeres after replication. That nucleoprotein structure is recombinogenic and depends on ATR, which RPA itself recruits.
- **Effect size:** strong — ATR inhibition disrupted the ALT process and caused chromosome fragmentation and apoptosis, with killing reported as selective for ALT-dependent cancer cells.
- **Variance:** vs telomerase-positive (non-ALT) cell lines; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Cell lines only, no in-vivo arm, and the central selectivity claim was not reproduced by Deeg 2016 or by the osteosarcoma panel in Goncalves 2020. Treat as the hypothesis that opened the field rather than as a settled vulnerability.

## Smith/Houghton (2015) — Clin Cancer Res

Reference: [PMID 25500058](#) · Type: preclinical · Inclusion: included

- **Intervention:** PARP inhibitor + temozolomide in sarcoma xenografts
- **Indication / model:** Pediatric Preclinical Testing Program in-vitro panel and xenografts, including 10 Ewing sarcoma models; talazoparib with temozolomide or topotecan
- **Design:** Trapping-competent PARP inhibition converts temozolomide base damage into replication-blocking lesions; pharmacodynamic loss of PARP appeared only in models that responded.
- **n:** 10 Ewing sarcoma xenograft models among a broader panel
- **Effect size:** moderate — Talazoparib potentiated temozolomide up to 85-fold in vitro, with the largest shifts in Ewing sarcoma and leukaemia lines. In vivo the combination was significantly synergistic in 5 of 10 Ewing xenografts and only modestly active in most other models, and both schedules represented the maximum tolerated combination.
- **Variance:** vs talazoparib alone, temozolomide alone, vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Ewing sarcoma, where PARPi sensitivity is driven by EWS-FLI1 rather than HR deficiency, so the mechanism is not the one proposed for uLMS. Talazoparib traps far more strongly than olaparib, making the potentiation an upper bound, and the myelosuppression that dominates the clinical regimen is exactly what the tolerability ceiling here predicts.

## Doebele/Low (2015) — Cancer Discov

Reference: [PMID 26216294](#) · Type: preclinical · Inclusion: included

- **Intervention:** LOXO-101 (larotrectinib) against an LMNA-NTRK1 fusion
- **Indication / model:** TRK-fusion-bearing human cancer cell lines in vitro and in vivo, alongside a soft-tissue sarcoma patient with lung metastases carrying an LMNA-NTRK1 fusion confirmed by in-situ proximity ligation

- **Design:** Selective ATP-competitive inhibition of TRKA/B/C. A fusion partner drives constitutive kinase activation, making the whole tumour dependent on that one signal, which is why response is fast when the fusion is genuinely the driver.
- **n:** n=1 patient; cell-line panel size not stated
- **Effect size:** strong — Fusion-bearing cell lines showed inhibition of the fusion oncoprotein with reduced proliferation in vitro and tumour growth in vivo. The sarcoma patient had rapid, substantial tumour regression with improvement in dyspnea and oxygen saturation.
- **Variance:** vs non-fusion cell lines for selectivity; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Preclinical content is thin next to the clinical case, and NTRK fusions are rare in leiomyosarcoma. Carried because the fusion has never been tested here and spindle-cell morphology with an earlier benign sign-out is the setting where an NTRK-rearranged mesenchymal tumour is missed.

## Judson/van der Graaf (2014) — Lancet Oncol

**Reference:** [PMID 24618336](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Doxorubicin + ifosfamide (intensified) vs doxorubicin alone (EORTC 62012)
- **Indication / model:** advanced or metastatic high-grade soft-tissue sarcoma, first line (1L); age 18-60, WHO PS 0-1; all histologies of high-grade STS, not LMS-specific
- **Design:** randomized controlled phase 3 (EORTC 62012), OS primary endpoint
- **n:** 455
- **Effect size:** 0.83 HR — median OS
- **Variance:** 95% CI 0.67–1.03; p=0.076
- **Toxicities:** febrile neutropenia G<sub>≥</sub>3 (103/224, 46%); leukopenia G<sub>≥</sub>3 (97/224, 43%); neutropenia G<sub>≥</sub>3 (93/224, 42%); anemia G<sub>≥</sub>3 (78/224, 35%); thrombocytopenia G<sub>≥</sub>3 (75/224, 33%)
- **Case fit:** partial
- **Notes:** The trial behind the rule that combination cytotoxics buy response and PFS but not OS in aggregate STS; the age band (18-60) matches her. Pooled STS, not uLMS-specific, so it is weaker evidence for her than LMS-04, which reversed the OS conclusion in histology-selected LMS.

## George/Krasner (2014) — Cancer

**Reference:** [PMID 24222211](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Letrozole in ER/PR-expressing uLMS
- **Indication / model:** unresectable uterine leiomyosarcoma with ER and/or PR expression by IHC (any); 27 patients, median 2 prior regimens (range 0-9); entry required ER and/or PR positivity — a population her ER/PR <1% tumor cannot join
- **Design:** single-arm phase 2, primary endpoint 12-week PFS
- **n:** 27
- **Effect size:** 50 % — PFS rate at 12 weeks
- **Variance:** 95% CI 30–67
- **Toxicities:** Reported as well tolerated; the dominant discontinuation reason was progression (85%). Per-term toxicity table sits behind the Wiley paywall.
- **Case fit:** none
- **Notes:** Included as the evidence that makes her ER <1%/PR <1% a genuine foreclosure: the only durable

letrozole benefit occurred in tumors expressing ER and PR in >90% of cells, the opposite end of the expression spectrum from hers. Toxicities empty after the fallback chain failed: abstract has no per-term data, no PMC deposit, Europe PMC subscription-only, Wiley page returns 402, and no trial registration is given in the abstract to reach a CT.gov AE tab.

## George/Muto (2014) — Cancer

**Reference:**PMID 24923260 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Intraperitoneal morcellation and outcomes of localized uLMS (DFCI/BWH)
- **Indication / model:** uterus-limited uLMS discovered at surgery for presumed leiomyoma (any); 58 women, 2007-2012: 19 morcellated vs 39 total abdominal hysterectomy at first surgery
- **Design:** retrospective two-institution cohort, multivariate-adjusted
- **n:** 58
- **Effect size:** 3.18 HR — recurrence-free survival, morcellated vs intact
- **Variance:** 95% CI 1.5–6.8; p=0.003
- **Toxicities:** Surgical-outcomes cohort; no systemic-therapy safety data.
- **Case fit:** partial
- **Notes:** Directly on the target\_validator's open question: if her September 2024 hysterectomy for presumed benign leiomyoma used morcellation, recurrence risk roughly triples and shifts toward peritoneal dissemination — which changes staging imaging (abdomen/pelvis surveillance), not just history-taking. The operative report is the cheap decisive document.

## Murai/Pommier (2014) — J Pharmacol Exp Ther

**Reference:**PMID 24650937 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP trapping as the reason olaparib pairs with temozolomide
- **Indication / model:** genetically modified chicken DT40 lymphoma lines with human DU145 prostate and SF295 glioblastoma cells; olaparib and veliparib against temozolomide, camptothecin, cisplatin and etoposide
- **Design:** PARP inhibitors kill in two separable ways: catalytic blockade, and trapping of PARP-DNA complexes at damage sites. Which one matters depends on the partner drug — temozolomide-induced N-methylpurine lesions recruit PARP1 to base-excision intermediates, where a trapped complex becomes the lethal lesion.
- **Effect size:** moderate — Temozolomide combinations needed trapping as well as catalytic inhibition, and olaparib, the stronger trapper, outperformed veliparib. Camptothecin combinations ran on catalytic inhibition alone, and cisplatin or etoposide gained little or nothing from either drug.
- **Variance:** vs veliparib, a potent catalytic inhibitor that traps poorly, as the mechanistic contrast to olaparib; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Avian and epithelial cell lines, no sarcoma and no in-vivo arm. Its value here is explaining why the uLMS trial regimen pairs olaparib with temozolomide specifically rather than with a platinum.

## Germano/Allavena (2013) — Cancer Cell

**Reference:**PMID 23410977 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin — mononuclear-phagocyte depletion

- **Indication / model:** four syngeneic mouse tumour models plus human blood monocytes; myeloid-cell transfer and depletion experiments in trabectedin-resistant tumours
- **Design:** Caspase-8 apoptosis triggered through TRAIL receptors. Monocytes carry signalling TRAIL receptors where neutrophils and lymphocytes carry decoys, which confines the killing to the mononuclear phagocyte compartment.
- **Effect size:** strong — Trabectedin killed monocytes and macrophages in blood, spleen and tumour while leaving other leukocytes intact, and tumour angiogenesis fell alongside them. Transfer and depletion experiments in trabectedin-resistant tumours placed the myeloid kill inside the antitumour effect rather than beside it. Monocyte counts, including tumour-associated macrophages, dropped in treated patients as well.
- **Variance:** vs vehicle-treated tumour-bearing mice; trabectedin-resistant tumour variants used to separate tumour-cell from myeloid effects; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** None of the four models is leiomyosarcoma, and the human confirmation is peripheral monocyte counts rather than sarcoma biopsies. Whether any of this reaches an infiltrative pulmonary metastasis is untested.

## Tran/Rosenberg (2013) — J Exp Med

**Reference:** [PMID 23712432](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** On-target toxicity of FAP-directed therapy
- **Indication / model:** FAP-reactive CAR T cells transferred into mice bearing a variety of subcutaneous tumours, across two mouse strains; FAP staining of 18 human tumours and of mouse and clinical-grade human multipotent bone marrow stromal cells
- **Design:** FAP is not tumour-restricted. It is expressed on PDGFR- $\alpha$ <sup>+</sup> Sca-1<sup>+</sup> multipotent bone marrow stromal cells in mice and on clinical-grade human BMSCs, so any FAP-directed effector reaches the marrow niche.
- **n:** 18 human tumours stained; two mouse strains tested
- **Effect size:** negative — FAP-reactive CAR T cells were only weakly active against tumours yet caused significant cachexia and lethal bone toxicity in two mouse strains. Both mouse and human multipotent BMSCs were recognised by the same T cells.
- **Variance:** vs untreated and control-CAR tumour-bearing mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** A CAR T-cell platform rather than a radioligand, and the dose delivered to marrow stroma by a peptide with renal clearance is not the same exposure. It remains the sharpest evidence that FAP positivity on a PET scan does not make FAP a tumour-exclusive target.

## Eisinger-Mathason/Simon (2013) — Cancer Discov

**Reference:** [PMID 23906982](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Intratumoural hypoxia, HIF-1 $\alpha$ /PLOD2 and sarcoma lung metastasis
- **Indication / model:** autochthonous LSL-KrasG12D/+; Trp53fl/fl murine undifferentiated pleomorphic sarcoma and allografts, with HIF-1 $\alpha$  or PLOD2 deletion and PLOD2 rescue; human sarcoma expression analysis
- **Design:** Hypoxia drives HIF-1 $\alpha$  to induce PLOD2, which modifies and organises collagen in the primary tumour. That altered matrix supports migration and dissemination without affecting primary tumour growth.
- **Effect size:** strong — Losing HIF-1 $\alpha$  or PLOD2 disrupted collagen modification, cell migration and pulmonary metastasis while leaving primary growth intact, and re-expressing PLOD2 restored metastatic potential in

HIF-1 $\alpha$ -deficient tumours. Human metastatic primary lesions carried elevated HIF1A and PLOD2, and pharmacologic PLOD inhibition suppressed metastases.

- **Variance:** vs HIF-1 $\alpha$ -proficient and PLOD2-proficient tumours; ectopic PLOD2 re-expression in HIF-1 $\alpha$ -deficient tumours; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** This is intratumoural hypoxia in a UPS model and speaks to lung tropism in sarcoma generally. It says nothing about ambient hypoxia at 2,600 m: no sarcoma model has been run under chronic hypobaric conditions, and the two forms of hypoxia are not interchangeable.

## van der Graaf/Hohenberger (2012) — Lancet

**Reference:** [PMID 22595799](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pazopanib (PALETTE)
- **Indication / model:** metastatic non-adipocytic soft-tissue sarcoma after standard chemotherapy (2L+); angiogenesis-inhibitor-naïve, prior anthracycline required; LMS was the largest histology; uterine sarcoma subset analyzed separately (Benson 2016 row)
- **Design:** randomized double-blind placebo-controlled phase 3, 2:1
- **n:** 369
- **Effect size:** 0.31 HR — median PFS
- **Variance:** 95% CI 0.24–0.4;  $p < 0.0001$
- **Toxicities:** fatigue (155/239, 65%); diarrhea (138/239, 58%); nausea (129/239, 54%); weight loss (115/239, 48%); hypertension (99/239, 41%)
- **Case fit:** partial
- **Notes:** Her label-approved later-line backbone once anthracycline exposure is behind her; PALETTE required prior chemotherapy, so it does not compete with LMS-04 now. Per-term G3-4 rates not in the abstract; Lancet full text paywalled and no PMC deposit — any-grade rates above are as published in the abstract.

## McClendon/Knudsen (2012) — Cell Cycle

**Reference:** [PMID 22751436](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CDK4/6 inhibition antagonising concurrent cytotoxic therapy
- **Indication / model:** RB-proficient and RB-deficient triple-negative breast cancer cell lines treated with palbociclib plus doxorubicin
- **Design:** Genotoxic cytotoxicity needs cells to cycle. An RB-mediated G1 arrest pulls them out of the phase where the damage is lethal, so the arrest shields the cell rather than adding to the kill.
- **Effect size:** negative — In RB-proficient models the pairing was cytostatically cooperative but protected cells from doxorubicin-mediated killing, and viable populations recurred after treatment. In RB-deficient cells palbociclib changed nothing about the doxorubicin response, confirming the protection runs through RB.
- **Variance:** vs doxorubicin alone; RB-deficient lines as the internal control for RB dependence; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Breast models and doxorubicin rather than gemcitabine, but gemcitabine is likewise S-phase-dependent. This is the mechanistic caution to raise against a concurrent abemaciclib-plus-gemcitabine schedule in an Rb-intact tumour.

## Burt/Jaklitsch (2011) — Ann Thorac Surg

Reference:PMID 21867989 · Type: clinical · Inclusion: included

- **Intervention:** Repeated pulmonary metastasectomy for leiomyosarcoma (Brigham series)
- **Indication / model:** pulmonary metastases from sarcoma; LMS subset analyzed (31/82) (any); single-institution 1989-2004; LMS patients were mostly women (77%), had fewer lesions (mean 1.9) and 48% underwent repeat metastasectomy
- **Design:** retrospective single-institution cohort
- **n:** 82
- **Effect size:** 70 months — median OS, LMS vs non-LMS metastasectomy
- **Variance:**  $p=0.049$
- **Toxicities:** Surgical complications not reported in the abstract.
- **Case fit:** partial
- **Notes:** Why the consulting oncologist keeps surgery on the table: resected LMS lung metastases can run an indolent course, and repeat resection is compatible with long survival.  $n=31$  LMS at one hospital, all selected for operability — her bilateral infiltrative pattern is exactly what these series screen out, so this is a best-case ceiling, not an expectation.

## Park/Nam (2011) — Gynecol Oncol

Reference:PMID 21565389 · Type: clinical · Inclusion: included

- **Intervention:** Tumor morcellation and prognosis in apparently early uLMS (Asan)
- **Indication / model:** stage I-II uterine leiomyosarcoma, morcellated vs intact at surgery (any); 56 consecutive patients 1989-2010: 25 with, 31 without tumor morcellation; groups balanced except smaller tumors in the morcellated group
- **Design:** retrospective single-institution comparative cohort
- **n:** 56
- **Effect size:** 3.11 OR — OS, morcellated vs intact (multivariate)
- **Variance:** 95% CI 1.07–9.06;  $p=0.038$
- **Toxicities:** Surgical-outcomes cohort; no systemic-therapy safety data.
- **Case fit:** partial
- **Notes:** Independent replication of the morcellation signal in a Korean series: tripled dissemination rate and worse OS. Small  $ns$  and retrospective design in both morcellation rows; the direction is consistent, the effect sizes are imprecise.

## Dean/Knudsen (2010) — Oncogene

Reference:PMID 20473330 · Type: preclinical · Inclusion: included

- **Intervention:** RB loss as the determinant of CDK4/6-inhibitor failure
- **Indication / model:** panel of breast cancer cell lines plus RB knockdown in immortalised mammary epithelium and multiple breast cancer subtypes; PD-0332991 (palbociclib)
- **Design:** CDK4/6 inhibition works by keeping RB active as a transcriptional repressor of E2F targets and by holding down CDK2 activity. Chronic RB loss moves cells into a CDK4/6-independent state; acquired resistance follows the same logic through loss of CDK2 restraint.
- **Effect size:** strong — Chronic RB loss produced CDK4/6 independence and resistance to palbociclib, and



direct RB knockdown reproduced it. Acquired resistance in initially sensitive models mapped to attenuated CDK2 inhibitors, so redundancy within the cell-cycle machinery is a second route to failure.

- **Variance:** vs RB-proficient parental lines and non-targeting shRNA controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Breast models throughout. Carried here because it supplies the mechanistic reason an Rb-intact requirement is a real eligibility gate rather than an enrichment preference.

## Hensley/Rose (2008) — Gynecol Oncol

**Reference:**PMID 18534250 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel, first-line metastatic uLMS (GOG 087L)
- **Indication / model:** metastatic uterine leiomyosarcoma, chemotherapy-naïve (1L); uLMS only; first-line; multicentre cooperative-group setting
- **Design:** single-arm phase 2 (Gynecologic Oncology Group)
- **n:** 42
- **Effect size:** 35.8 % — ORR
- **Variance:** 95% CI 23.5–49.6
- **Toxicities:** neutropenia G3 (n=42, 5%); neutropenia G4 (n=42, 12%); anemia G3 (n=42, 24%); thrombocytopenia G3 (n=42, 9.5%); fatigue G3 (n=42, 17%); pulmonary toxicity G4 (1/42, 2.4%)
- **Case fit:** strong
- **Notes:** The uLMS-specific first-line estimate for the fallback regimen: ORR 35.8%, mPFS 4.4 months — respectable but well short of LMS-04's 12.2-month PFS. Single-arm, n=42; no comparator.

## Aune/Pommier (2008) — Clin Cancer Res

**Reference:**PMID 18927284 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin — TC-NER- and VHL-dependent RNA Pol II degradation
- **Indication / model:** isogenic repair-mutant human fibroblasts (CSB, XPA, XPC, XPD, XPF, XPG) and VHL-deficient cells, with complemented counterparts; several cancer cell lines
- **Design:** Trabectedin adducts recruit transcription-coupled nucleotide excision repair, and the stalled complex is resolved by proteasomal degradation of elongating RNA Pol II, which needs both TC-NER and an intact VHL complex.
- **Effect size:** strong — Transcribing RNA Pol II was degraded rapidly after trabectedin exposure, and cells defective in TC-NER or VHL failed to degrade it. VHL-deficient cells also resisted trabectedin killing, while XPC cells, defective only in global-genome repair, behaved like wild type.
- **Variance:** vs repair-proficient parental lines and genetically complemented mutants; proteasome inhibition with MG132; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Repair-mutant fibroblasts and epithelial lines, no sarcoma model. This is the reason trabectedin sensitivity tracks repair proficiency rather than proliferation rate, which is what separates it from a conventional cytotoxic.

## Allavena/D'Incalci (2005) — Cancer Res



**Reference:**PMID 15805300 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trabectedin — macrophage differentiation and CCL2/IL-6 output
- **Indication / model:** human blood monocytes, M-CSF-differentiated macrophages, and tumour-associated macrophages freshly isolated from ovarian cancer; monocyte counts in treated patients
- **Design:** DNA minor-groove binding with selective interference in inducible transcription; monocytes die at low nanomolar exposure, and below that concentration the drug shuts down CCL2 and IL-6 transcription.
- **Effect size:** moderate — Monocytes underwent apoptosis at 5 nmol/L while lymphocytes tolerated roughly five-fold more drug. Sub-cytotoxic exposure blocked monocyte-to-macrophage differentiation and cut CCL2 and IL-6 output from monocytes, macrophages, ovarian TAMs and tumour cells alike.
- **Variance:** vs untreated monocytes/macrophages; lymphocytes as the comparator leukocyte population; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Human cells in vitro plus ovarian TAMs; no sarcoma tissue anywhere in the paper. CCL2 suppression is the proposed link to reduced monocyte recruitment, inferred rather than measured in tumour.

### Meco/Riccardi (2003) — Cancer Chemother Pharmacol

**Reference:**PMID 12783202 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ET-743 (trabectedin) + doxorubicin — preclinical basis for the combination
- **Indication / model:** TE-671 rhabdomyosarcoma clonogenic assay and nude-mouse xenograft; C3H mice bearing UV2237M fibrosarcoma and its P-gp-overexpressing doxorubicin-resistant subline
- **Design:** Two non-overlapping DNA-damage mechanisms — minor-groove alkylation whose cytotoxicity runs through transcription-coupled repair, versus topoisomerase II poisoning — combined without any pharmacokinetic interaction to account for the gain.
- **Effect size:** moderate — Isobologram analysis gave combination indices just under 1, so the in-vitro interaction is additive at best. In vivo the pair produced log cell kill of 0.85-1.12 in TE-671 where either agent alone reached only 0.13-0.33, and the combination worked in a P-gp-positive subline resistant to both single drugs. Giving ET-743 one hour ahead edged out simultaneous or reversed dosing (LCK 1.12 vs 0.85 and 0.92), and doxorubicin levels in tumour, liver, heart and kidney were unchanged by co-administration.
- **Variance:** vs each drug alone and untreated controls; pharmacokinetic sampling in both single-agent and combination arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Rhabdomyosarcoma and fibrosarcoma models from 2003; no leiomyosarcoma and no uterine model. The sequencing advantage rests on one small in-vivo comparison, and nothing in this or any other preclinical study addresses the trabectedin maintenance phase, which is an empirical clinical design.

### Kusmartsev/Gabrilovich (2003) — Cancer Res

**Reference:**PMID 12907617 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATRA differentiating immature myeloid suppressor cells
- **Indication / model:** multiple tumour-bearing mouse models with adoptive transfer of immature myeloid cells; ATRA combined with two cancer vaccines
- **Design:** ATRA drives immature myeloid cells to differentiate into dendritic cells, macrophages and

granulocytes, which removes the suppressive population rather than killing tumour cells directly.

- **Effect size:** moderate — ATRA sharply reduced immature myeloid cells across every model tested, and adoptive transfer confirmed the cells had differentiated rather than died. CD4 and CD8 tumour-specific responses improved and vaccine effects lasted longer, with no direct antitumour activity of ATRA itself.
- **Variance:** vs vaccine alone and untreated tumour-bearing mice; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross\_tumor\_only
- **Notes:** Vaccine-era mouse models with no sarcoma and no checkpoint blockade; the trial pairs ATRA with cemiplimab, a combination this work predates. Doi not indexed in PubMed for this record, so the identifier is left null rather than guessed.

## Hensley/Spriggs (2002) — J Clin Oncol

**Reference:**PMID 12065559 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemcitabine + docetaxel in leiomyosarcoma (MSKCC phase 2)
- **Indication / model:** unresectable leiomyosarcoma (29/34 uterine), 0-2 prior regimens (any); mixed-line: 18 chemo-naive, 16 doxorubicin-pretreated; 85% uterine primaries
- **Design:** single-arm phase 2 with pharmacokinetic substudy
- **n:** 34
- **Effect size:** 53 % — ORR
- **Variance:** 95% CI 35–70
- **Toxicities:** neutropenia G3 (n=34, 15%); neutropenia G4 (n=34, 6%); thrombocytopenia G3 (n=34, 26%); thrombocytopenia G4 (n=34, 3%); febrile neutropenia G $\geq$ 3 (n=34, 6%)
- **Case fit:** partial
- **Notes:** The 53% ORR that made gem-doce a uLMS standard is from 34 patients at one centre with mixed lines of therapy; the GOG first-line trial (PMID 18534250) halved that estimate. Read the two together before quoting a response rate.

## Van Glabbeke/Tursz (1999) — J Clin Oncol

**Reference:**PMID 10458228 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Prognostic factors for first-line anthracycline chemotherapy in advanced STS (EORTC pooled)
- **Indication / model:** advanced soft-tissue sarcoma treated with anthracycline-containing first-line regimens (1L); 2,185 patients pooled from seven EORTC trials; LMS included among histologies
- **Design:** pooled prognostic-factor analysis of prospective trial data (Cox and logistic models)
- **n:** 2185
- **Effect size:** median OS 51 weeks; ORR 26% mixed — independent prognostic factors for OS
- **Toxicities:** Prognostic analysis; no treatment-safety data.
- **Case fit:** partial
- **Notes:** The trial-grade basis for reading her ~23-month interval as favorable: a long time lapse between initial diagnosis and advanced-disease treatment independently predicted longer survival in 2,185 first-line anthracycline patients. Her interval is measured from an unrecognised resected primary, an unusual variant of that variable, and her young age and (assumed) ECOG 1 sit on the favorable side of the same model.

## Pastorino/Putnam (1997) — J Thorac Cardiovasc Surg

**Reference:**PMID 9011700 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pulmonary metastasectomy (International Registry of Lung Metastases)
- **Indication / model:** resectable lung metastases from solid tumors; sarcoma was 42% of the registry (2,173/5,206) (any); 5,206 metastasectomies from 18 thoracic-surgery units; 88% complete resections; surgical candidates only — a fitness- and biology-selected population
- **Design:** international multi-institutional registry, retrospective prognostic analysis
- **n:** 5206
- **Effect size:** 36 % — 5-year OS after complete metastasectomy
- **Toxicities:** Perioperative morbidity and mortality not reported in the abstract; pre-CTCAE-era surgical registry.
- **Case fit:** partial
- **Notes:** The reference dataset behind offering lung surgery at all, and the source of the prognostic strata (completeness, DFI, lesion number) her team would apply. Entirely retrospective and surgeon-selected: the 5-year figures describe patients healthy and oligometastatic enough to be operated on, not bilateral infiltrative disease like hers unless induction converts it.